CUSTOMER CHURN PREDICTION & AGENTIC RETENTION STRATEGY

From Predictive Analytics to Intelligent Intervention

✦ MILESTONE 2 ✦

Team Member	Role
Vanshika Yadav	ML Modeling · Business Context · Input-Output Design · Structured Output · Code Development
Ronit Singh	LangGraph Workflow · State Management · System Architecture · Agent Workflow Docs · Code
Riya Garg	EDA · Visualization · UI Development · Deployment · README Documentation
Sankalp	RAG (FAISS) · Prompt Engineering · Testing · Model Evaluation · Code Development

Faculty Advisor	Prof. Bipul Shahi
Course	Gen-AI
Group	Team RetainX AI

ABSTRACT

This project presents the design and implementation of an advanced AI-driven customer churn intelligence system that evolves from a traditional predictive analytics model into a fully agentic AI-powered decision support system.

In Milestone 1, a machine learning pipeline was developed to predict customer churn using structured telecom data. However, prediction alone does not solve the business problem of customer retention. Therefore, in Milestone 2, the system was extended into an Agentic AI architecture capable of reasoning about churn risk, retrieving relevant retention strategies, planning interventions, and generating structured recommendations.

The system integrates machine learning, retrieval-augmented generation (RAG), and large language models (LLMs) using a LangGraph-based workflow. This enables the system to move beyond prediction and provide actionable business insights, making it a complete intelligent retention assistant.

PROBLEM STATEMENT & BUSINESS CONTEXT

Customer churn is one of the most critical challenges faced by subscription-based businesses such as telecom companies. Losing customers directly impacts revenue, customer lifetime value, and growth potential.

Traditional systems can predict churn but fail to provide actionable insights on how to prevent it. Businesses require not only identification of at-risk customers but also intelligent strategies to retain them.

Key Challenges Addressed

- Identifying high-risk customers
- Understanding the reasons behind churn
- Recommending effective retention strategies
- Reducing reliance on manual decision-making

This project aims to bridge the gap between prediction and action by introducing an agentic AI system that can autonomously analyze, reason, and recommend.

OBJECTIVES

The primary objectives of Milestone 2 are:

- Extend the ML-based churn prediction system into an intelligent AI agent
- Implement a multi-step agent workflow using LangGraph
- Integrate Retrieval-Augmented Generation (RAG) for strategy retrieval
- Enable structured and explainable output generation
- Reduce hallucinations through prompt engineering

- Build a deployable application for real-world use

SYSTEM OVERVIEW

Milestone 1: ML-Based Prediction	Milestone 2: Agentic AI System
• Data preprocessing • Model training (XGBoost) • Evaluation and threshold tuning • Prediction of churn probability	• Risk reasoning using model output • RAG-based strategy retrieval • LangGraph workflow execution • LLM-based structured output generation

DATASET INFORMATION

Dataset: Telco Customer Churn Dataset | **Total Records:** 7,043 | **Features:** 20 input variables + 1 target variable

The dataset contained 11 missing values in the *TotalCharges* column. These entries corresponded to customers with zero tenure. Since they represented a very small portion of the dataset, these rows were removed to maintain data consistency.

Target Variable

• **Churn** (Yes / No)

Feature Categories

Category	Features
1. Customer Demographics	Gender, SeniorCitizen, Partner, Dependents
2. Services Subscribed	PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies
3. Account Information	Tenure, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges

EXPLORATORY DATA ANALYSIS (EDA) & INSIGHTS

Exploratory Data Analysis was conducted to understand customer behavior patterns, identify key churn drivers, and support both machine learning modeling and agentic reasoning in the system. The analysis combines statistical summaries and visual insights derived from the Telco Customer Churn dataset.

1. Churn Distribution

The dataset shows that approximately **26.6%** of customers have churned, while **73.4%** have been retained.

Figure 1: Churn Distribution (26.6% Churn vs 73.4% Retained)

Insight	
• The dataset is moderately imbalanced, with a majority of customers not churning.	• A churn rate of ~26% is significant and indicates a serious business concern.
Business Interpretation	
• Even a small percentage of churn can result in large revenue losses.	• This justifies the need for predictive modeling and retention strategies.

2. Churn Rate by Contract Type

Contract Type	Churn Rate
Month-to-month	**~42.7%**
One-year	**~11.3%**

Two-year	**~2.8%**

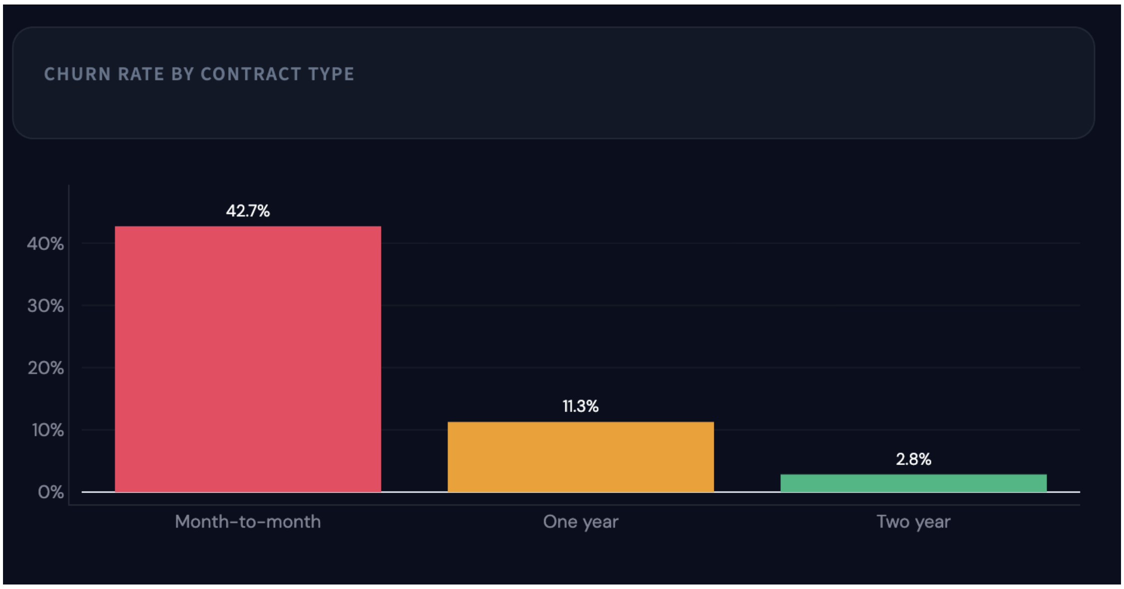

Figure 2: Churn Rate by Contract Type

Insight: Customers with month-to-month contracts have the highest churn rate. Long-term contracts (especially two-year plans) show very low churn.

Business Interpretation: Lack of commitment in short-term plans leads to higher churn. Long-term contracts act as a strong retention mechanism.

3. Churn Rate by Internet Service

• **Fiber optic:** ~41.9% churn • **DSL:** ~19.0% churn • **No internet:** ~7.4% churn

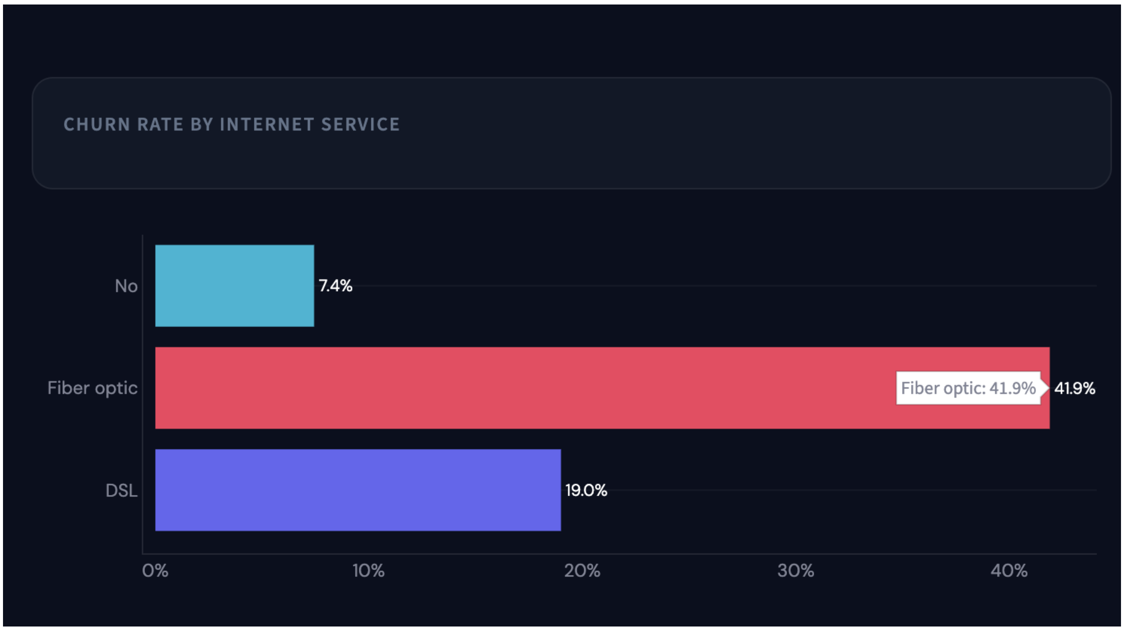

Figure 3: Churn Rate by Internet Service Type

Insight: Customers using fiber optic internet show the highest churn. DSL users churn less, while customers without internet services have minimal churn.

Business Interpretation: Fiber optic users may face higher costs or service dissatisfaction, indicating a need for service quality improvement or pricing optimization.

4. Churn Rate by Payment Method

• **Electronic check:** ~45.3% churn • **Bank transfer:** ~16.7% • **Credit card:** ~15.3% • **Mailed check:** ~19.2%

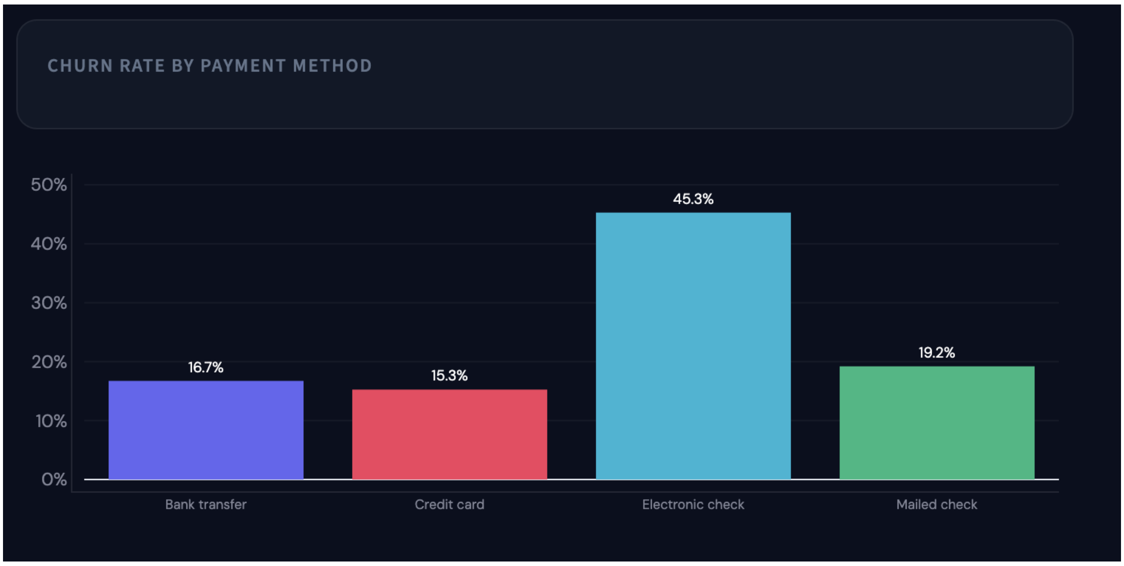

Figure 4: Churn Rate by Payment Method

Insight: Customers using electronic check have the highest churn rate.

Business Interpretation: This group may represent less committed users with higher friction in the payment experience — a strong signal for targeted retention strategies.

5. Tenure Distribution vs Churn

The tenure distribution shows that customers with **low tenure (0–12 months)** have the highest churn, while long-tenure customers are significantly more stable.

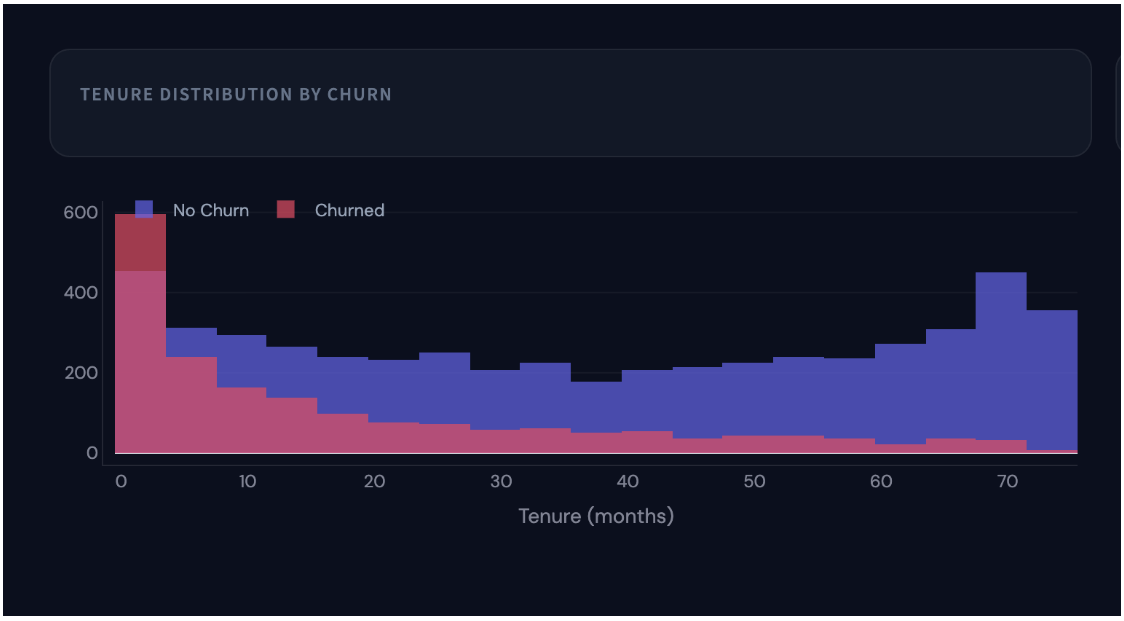

Figure 5: Tenure Distribution by Churn Status

Insight: Early-stage customers are at the highest risk of churn.

Business Interpretation: The first few months are critical for retention. Onboarding experience plays a key role.

6. Monthly Charges vs Churn

From the box plot, churned customers tend to have **higher monthly charges**, while non-churned customers are more concentrated at lower price ranges.

Figure 6: Monthly Charges vs Churn (Box Plot)

Insight: Higher pricing correlates with increased churn risk.

Business Interpretation: Customers may perceive low value for high cost. Pricing strategy directly impacts retention.

7. Key Business Insights (Derived)

Key Insight	Business Finding
1. Month-to-Month Contract Risk	Customers with monthly contracts churn at ~43%, nearly 3× higher than long-term customers.
2. Electronic Check Payment Risk	Electronic check users show ~45% churn, the highest among all payment methods.
3. Low Tenure = High Risk	Customers with tenure less than 12 months are significantly more likely to churn.
4. Long-Term Contracts Retain Customers	Two-year contract customers show <5% churn, indicating strong retention power.
5. Lack of Tech Support Increases Churn	Customers without tech support show significantly higher churn compared to those with support.
6. High Monthly Charges Drive Churn	Customers paying above ~$70/month show elevated churn risk, especially without bundled services.

8. Impact on System Design

Machine Learning Model	Agentic AI System
• Feature selection (tenure, contract, charges, etc.) • Threshold tuning • Focus on recall	• Risk reasoning logic (low tenure, high charges) • RAG query generation • Strategy recommendations

Conclusion of EDA: The exploratory analysis clearly highlights that churn is not random but driven by identifiable behavioral and financial patterns. These insights form the foundation for both predictive modeling and intelligent retention strategy generation in the system.

SYSTEM ARCHITECTURE (AGENTIC AI)

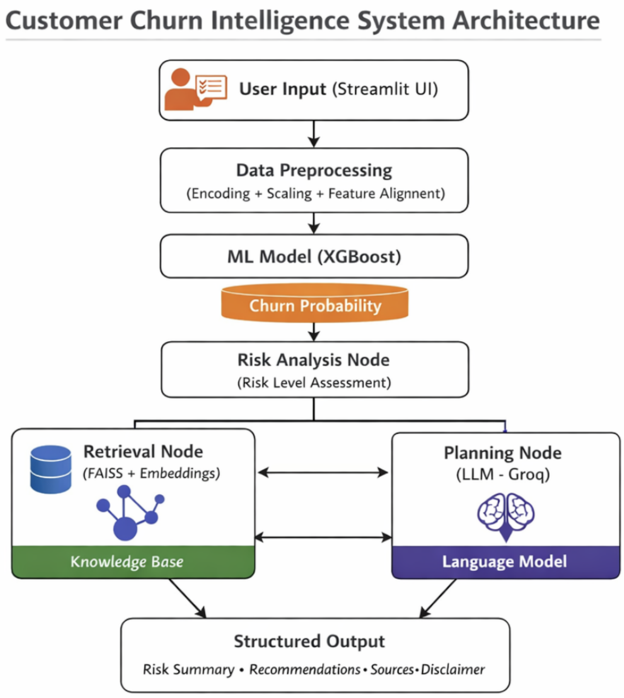

Figure 7: Customer Churn Intelligence System Architecture

AGENTIC AI IMPLEMENTATION

1. Framework: LangGraph (Workflow & State)

The core of the agentic system is implemented using LangGraph, which enables the creation of stateful, multi-step workflows.

LangGraph allows:

- Sequential execution of nodes
- Explicit state passing
- Modular design of AI reasoning steps

Workflow Structure:

```
risk_node → retrieval_node → planning_node
```

Each node performs a specialized task and updates the shared state.

2. State Management

The system uses explicit state management to ensure consistency across all steps.

```
class AgentState(TypedDict):
churn_prob: float
tenure: int
monthly: float
risk_level: str
reasons: List[str]
strategies: List[str]
sources: List[str]
final_output: str
```

- Maintains continuity across workflow
- Enables explainability
- Ensures structured processing
- Avoids loss of context

3. Risk Analysis Node

This is the first step in the agent pipeline. It converts churn probability into risk levels and identifies reasons behind churn.

Risk Level	Condition
■ **High Risk**	Probability > 0.7
■ **Medium Risk**	Probability > 0.4
■ **Low Risk**	Otherwise

Example reasoning: Low tenure → high churn | High monthly charges → dissatisfaction

4. RAG Implementation (FAISS)

What is RAG? Retrieval-Augmented Generation allows the model to retrieve relevant knowledge instead of generating from scratch.

Knowledge Base — stored in *retention_knowledge.json* — contains: Conditions, Strategies, and Sources.

1. Convert JSON into documents
2. Generate embeddings using: all-MiniLM-L6-v2
3. Store in FAISS vector database
4. Perform similarity search

Retrieval Logic:

```
results = vectorstore.similarity_search(query, k=3)
```

Benefits of RAG:

- Reduces hallucination
- Improves accuracy
- Ensures grounded recommendations
- Provides traceable sources

5. Planning Node (LLM Reasoning)

Model Used: Groq API (LLaMA 3.3)

Prompt Design — The prompt enforces strict constraints:

- Use ONLY retrieved strategies
- Do NOT generate new strategies
- Return structured JSON
- Provide explanations

Output Format (JSON):

```
{
"risk_summary": "...",
"recommendations": [...],
"sources": [...],
"business_impact": "...",
"disclaimer": "..."
}
```

Example Structured Output:

```
{
"risk_summary": "Customer has high churn risk due to low tenure and high charges",
"recommendations": [
"Offer discounted long-term contract",
"Provide personalized onboarding support"
],
"sources": ["Retention Strategy DB"],
"business_impact": "Customer likely to churn if no action is taken",
"disclaimer": "Prediction is probabilistic"
}
```

Anti-Hallucination Strategy:

- Grounded inputs (RAG)

- Strict prompt rules
- JSON-only output
- Parsing validation

6. Structured Output Generation

The final output is highly structured and business-ready, containing the following components:

Component	Description
Risk Summary	Overview of the customer's churn risk
Recommendations	Specific retention strategies
Sources	Traceable references from knowledge base
Business Impact	What happens if no action is taken
Disclaimer	Probabilistic nature of prediction

WORKING APPLICATION (STREAMLIT UI)

The system is deployed as an interactive web application with the following features:

- Dashboard with insights
- Customer input form
- Real-time churn prediction
- AI-powered recommendations
- Risk visualization

UI Flow:

1. User enters customer data
2. Model predicts churn
3. Agent workflow executes
4. Results displayed

Input–Output Specification

Input	Output
• Customer profile • Tenure • Charges	• Churn probability • Risk level • Retention strategies • Structured report

UI Screenshots

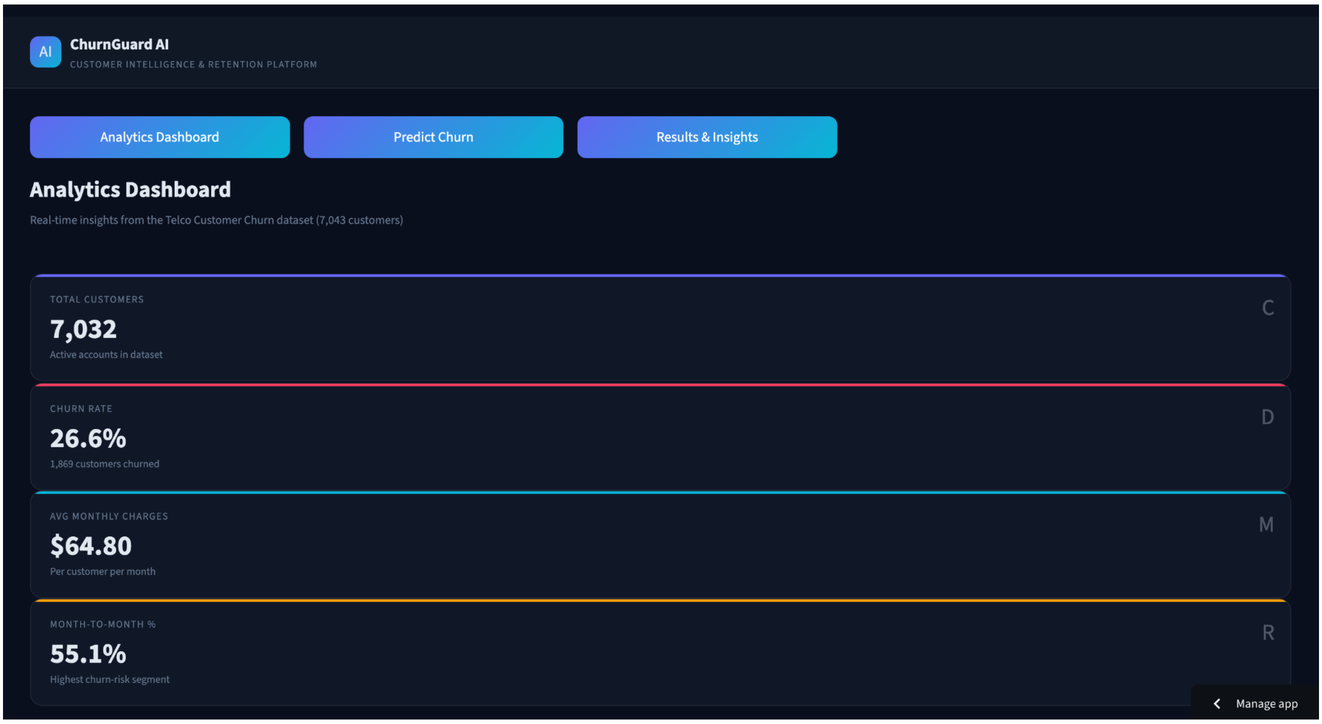

Figure 8: Analytics Dashboard — Key Metrics Overview

Figure 9: Analytics Dashboard — Churn Distribution & Contract Analysis

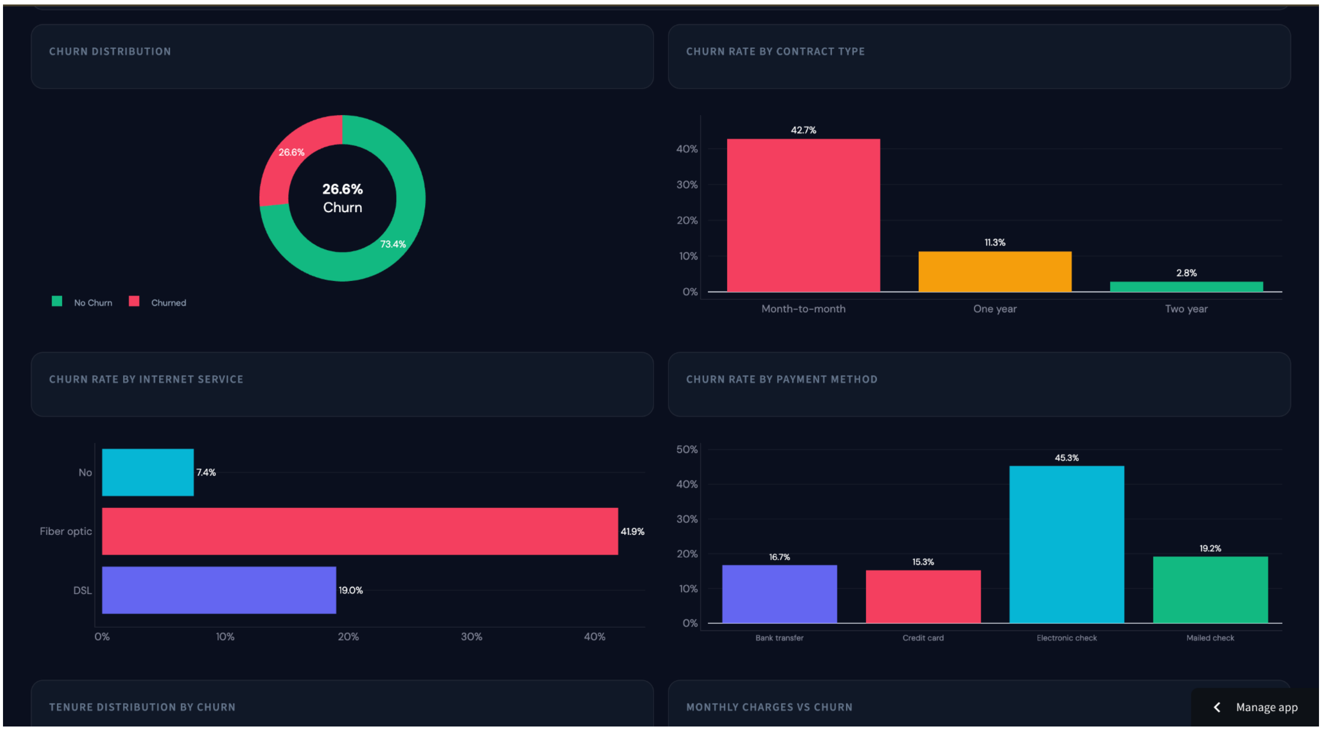

Figure 10: Analytics Dashboard — Internet Service & Payment Method Charts

Figure 11: Analytics Dashboard — Tenure Distribution & Monthly Charges

Figure 12: Analytics Dashboard — Key Business Insights Panel

Figure 13: Predict Churn — Customer Demographics Input Form

Figure 14: Predict Churn — Services Subscribed Input Form

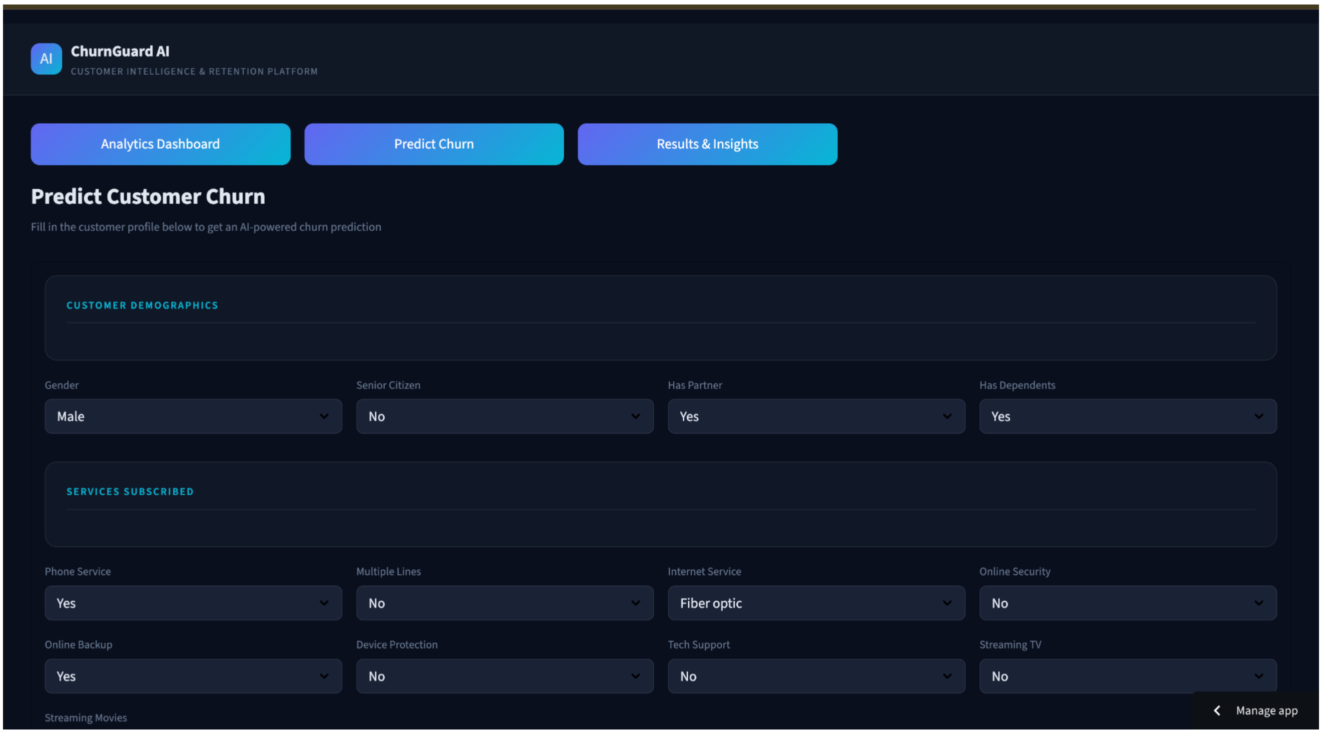

Figure 15: Results & Insights — 94% Churn Probability, Risk Factors & Recommendations

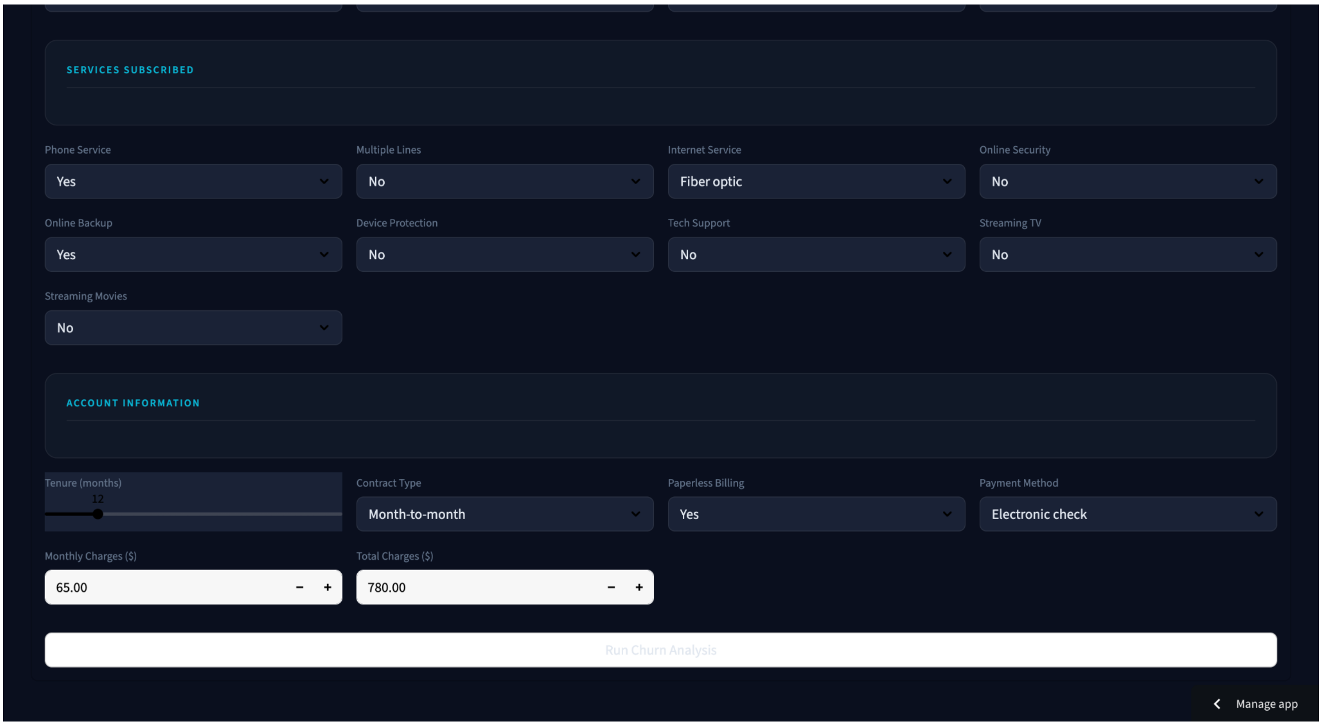

Figure 16: Results & Insights — Detailed Risk Factors

Figure 17: Results & Insights — Full Recommendations & Customer Profile

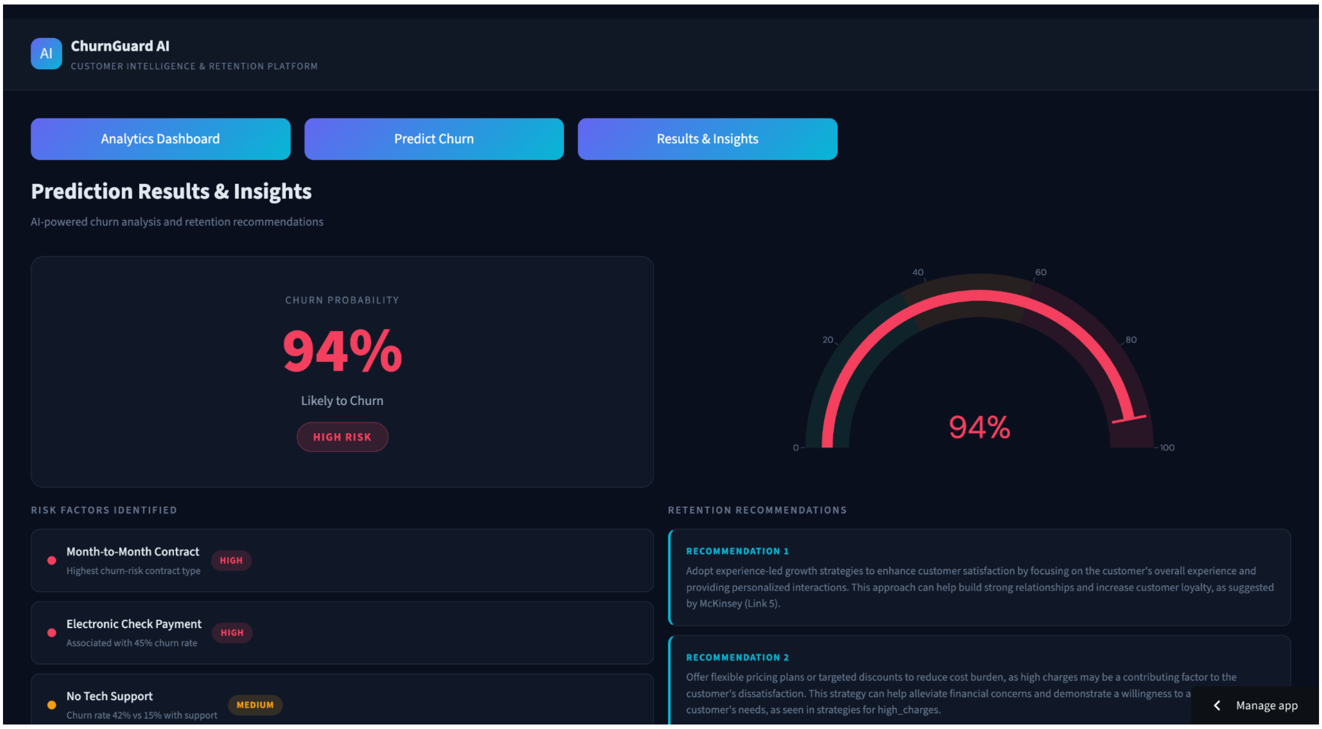

Figure 18: Results & Insights — Customer Profile Summary

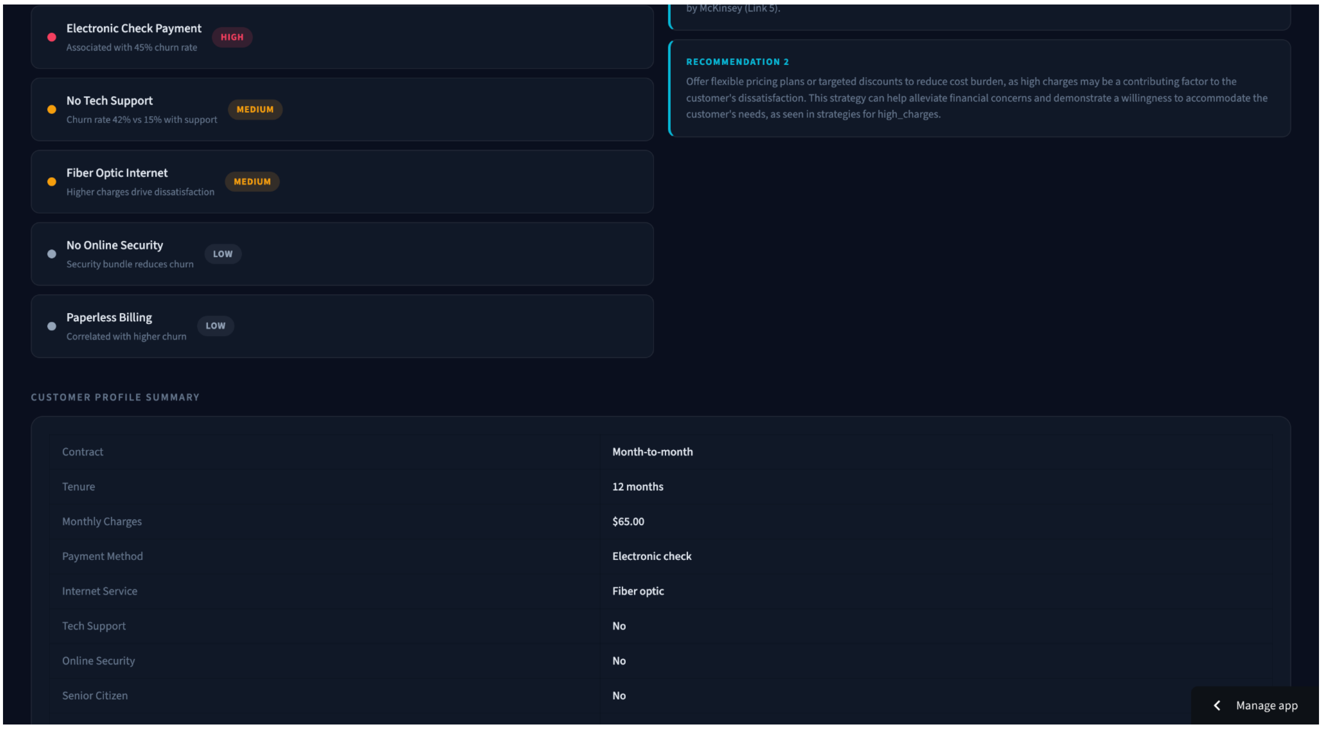

Figure 19: Results & Insights — Final Structured Output

MODEL PERFORMANCE INTEGRATION

The ML model (XGBoost) is integrated into the agent pipeline as follows:

- Provides churn probability as a float
- Acts as the primary input to the agentic reasoning workflow
- Enables contextual risk reasoning based on model output

DEPLOYMENT

Aspect	Details
Platform	Streamlit Cloud
Integration	GitHub integration with automatic deployment
Configuration	Environment variables managed via Secrets

CHALLENGES FACED

- API key handling across environments
- Dependency version conflicts during deployment
- RAG tuning for optimal retrieval quality
- JSON parsing errors in LLM responses
- Deployment debugging on Streamlit Cloud

FUTURE SCOPE

- Multi-agent systems for more complex decision-making
- Real-time streaming data integration
- Personalized retention strategies per customer segment
- SHAP explainability for model transparency
- Advanced LLM integration with fine-tuned models

TEAM CONTRIBUTIONS (MILESTONE 2)

Vanshika Yadav

- Led Machine Learning Modeling and ensured seamless integration with the agentic system
- Defined the business context and translated problem requirements into a technical solution
- Designed the input–output pipeline ensuring structured flow between ML and agent layers
- Developed the structured output format for the final AI-generated retention reports

Riya Garg

- Conducted Exploratory Data Analysis (EDA) and generated key business insights
- Created all visualizations and dashboard components for better data interpretation
- Developed and designed the complete UI using Streamlit
- Managed deployment of the application on Streamlit Cloud
- Maintained and structured the README documentation

Sankalp

- Implemented RAG (Retrieval-Augmented Generation) using FAISS
- Designed and optimized prompt engineering strategies to reduce hallucinations
- Performed testing and validation of the agentic pipeline
- Contributed to model evaluation and performance analysis
- Prepared the project report documentation

Ronit Singh

- Designed and implemented the LangGraph workflow for the agentic system
- Managed state handling and data flow across different nodes
- Developed the system architecture integrating ML, RAG, and LLM components
- Created the Agent Workflow Documentation
- Contributed to core code development and integration

Overall — All team members collaborated closely on: Integrating ML with the agentic AI pipeline, debugging and optimizing system performance, and ensuring successful deployment and end-to-end functionality.

CONTRIBUTION MATRIX & DELIVERABLES

Member	ML Model	Business Context	I/O Design	Struct. Output	LangGraph	State Mgmt	RAG	Prompt Engg	UI Dev	Deploy	Testing	Docs
Vanshika	✓	✓	✓	✓	✓	—	✓	—	—	—	—	—
Ronit	—	✓	—	—	✓	✓	✓	—	—	—	✓	✓
Riya	—	—	✓	—	—	—	✓	✓	✓	✓	—	✓

Sankalp	✓	—	—	✓	—	—	✓	✓	—	—	✓	✓

MILESTONE DELIVERABLES

Milestone 1 Deliverables — Completed

✓ Business understanding and problem framing

✓ Data cleaning and preprocessing pipeline

✓ Exploratory Data Analysis and visualization suite

✓ Multi-model comparison (Logistic Regression, Decision Tree, XGBoost)

✓ Feature engineering and encoding of categorical variables

✓ Threshold optimization for business-aligned recall improvement

✓ Churn probability prediction and confidence scoring

✓ Model persistence using joblib artifacts (model, scaler, encoders)

✓ Performance evaluation using accuracy, precision, recall, F1-score, and confusion matrix

✓ Development of an interactive Streamlit-based UI for predictions

✓ Deployment of the ML model as a live application on Streamlit Cloud

Milestone 2 — Agentic Extension — Completed

✓ Implementation of LangGraph-based agent workflow for autonomous decision-making

✓ Design of multi-step agent pipeline (Risk Analysis → Retrieval → Planning)

✓ Integration of ML model output with agentic reasoning system

✓ Implementation of RAG pipeline using FAISS for retrieval of retention strategies

✓ Creation of structured knowledge base (retention_knowledge.json) for grounded responses

✓ Embedding generation using sentence-transformers (MiniLM model)

✓ Similarity-based retrieval of top relevant strategies using vector search

✓ Prompt engineering to enforce structured, reliable, and hallucination-free outputs

✓ Integration of LLM (Groq / LLaMA) for reasoning and recommendation generation

✓ Structured output generation (Risk Summary, Recommendations, Sources, Business Impact, Disclaimer)

✓ Explicit state management across workflow using LangGraph state design

✓ Development of end-to-end Streamlit UI integrating ML + Agentic AI

✓ Deployment of complete system on Streamlit Cloud with environment variable management

✓ Testing and validation of agent outputs for correctness and consistency

✓ Documentation of system architecture and agent workflow

CONCLUSION

This project successfully evolves from a traditional machine learning-based churn prediction system into an advanced agentic AI solution. While the initial model effectively identifies customers at risk, the extended system goes a step further by autonomously analyzing churn factors and generating actionable retention strategies.

By integrating LangGraph workflows, RAG-based retrieval, and LLM-driven reasoning, the system provides structured, explainable, and business-oriented recommendations. This transforms the solution from a predictive tool into a decision-support system.

Overall, the project demonstrates how AI can move beyond prediction to intelligent intervention, helping businesses proactively reduce customer churn and improve retention outcomes.

REFERENCES

- Kaggle Dataset — Telco Customer Churn
- Scikit-learn — Machine Learning Library
- XGBoost — Gradient Boosting Framework
- LangGraph — Workflow Orchestration for LLMs
- FAISS — Facebook AI Similarity Search
- HuggingFace — Transformers & Embeddings
- Groq API — LLM Inference Platform

PROJECT ARTIFACTS

Resource	Link
GitHub Repository	https://github.com/connectwithvanshika/customer_churn_prediction
Streamlit Live App	https://customerchurnprediction-2k327gcbblu4dawrhawsit.streamlit.app/
Kaggle Dataset	https://www.kaggle.com/datasets/blastchar/telco-customer-churn
Video Explanation	https://drive.google.com/drive/u/1/folders/12rKe4vnmFKiYhrQIUKWCfCnlRTNMNwLt

RetainX AI — Intelligent Customer Retention Through Agentic AI